6.5 Combined Maneuvers

- For real-world missions, it is often necessary to change the size (a), shape (e) and inclination (i) of an orbit together.
  - When possible, it is better to combine the transfers into a single maneuver.
  - Actually reduces the overall Δv, the number of separate burns and the time needed to complete the operation.

- Assumption
  - Transfers originate from Circular Orbits
    - Elliptic Orbits will be studied in the frame of Lambert's algorithm
  - Burns occur at Nodal Crossings (like inclination-only maneuvers)

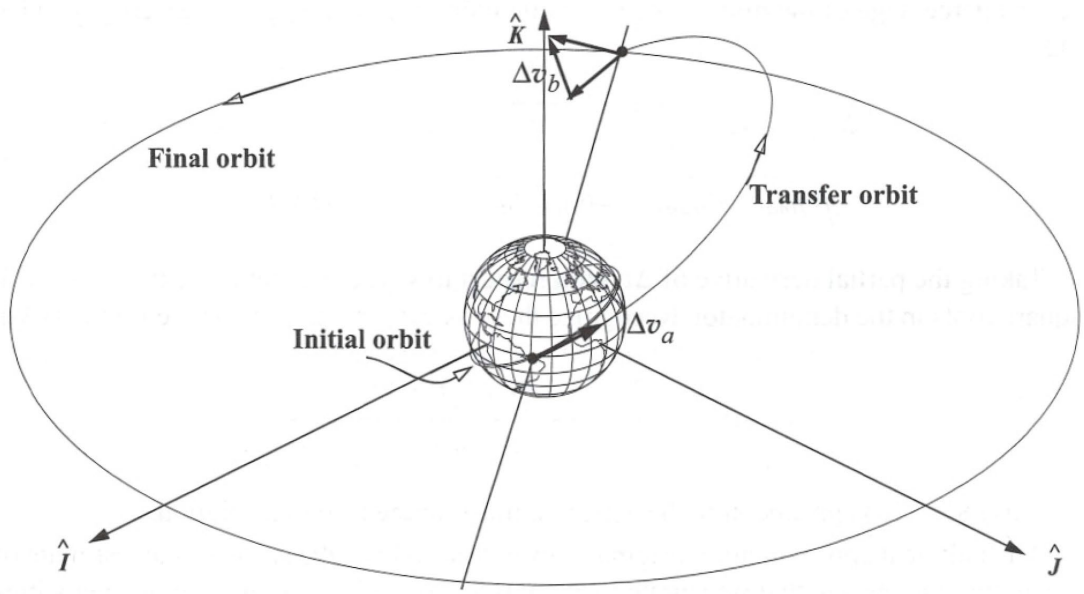


Figure 6-13. Geometry for a Combined Maneuver. A transfer from LEO to GEO is shown to the correct scale. The transfer orbit is just slightly below the initial 28.5° LEO orbit. Notice how close the LEO satellite is to the Earth when it's drawn to scale.

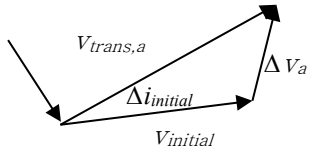
6.5.1 Minimum-Inclination Maneuvers

- Refers to a maneuver which minimizes total change in velocity (ΔV) for a given inclination change (Δi) with two burns, i.e.,

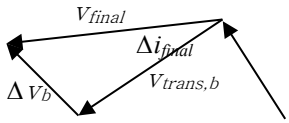
  Minimize $\Delta v = \Delta v(\Delta i_{initial}, \Delta i_{final})$

  Subject to $\Delta i = \Delta i_{initial} + \Delta i_{final}$

- Each velocity contains the Hohmann transfer contributions in $v_{trans,a}$, v_{final}.



$$\Delta v_a = \sqrt{v_{trans,a}^2 + v_{initial}^2 - 2v_{trans,a}v_{initial}\cos(\Delta i_{initial})}$$



$$\Delta v_b = \sqrt{v_{trans,b}^2 + v_{final}^2 - 2v_{trans,b}v_{final}\cos(\Delta i_{final})}$$

- Minimize the total change in velocity $\Delta v = \left|\Delta v_a\right| + \left|\Delta v_b\right|$

$\Delta i_{initial} = s\Delta i$, $\Delta i_{final} = (1-s)\Delta i$, *s:scale factor,*

Subject to *remember that Δi is given*

$\Leftrightarrow \Delta i = \Delta i_{initial} + \Delta i_{final}$

- Solution by Iteration

$$\frac{\partial \Delta v}{\partial s} = 0 \quad \Rightarrow \quad \sin(s\Delta i) = \frac{\Delta v_a v_{final} v_{trans,b} \sin((1-s)\Delta i)}{\Delta v_b v_{initial} v_{trans,a}} \qquad (1)$$

Because $\sin(s\Delta i)$ appears on both sides, we must iterate to find a solution:

I. Given $\left(\Delta i,\ v_{initial}, v_{trans,a}, v_{trans,b}, v_{final}\right)$

II. Guess s and compute $\left(\Delta v_a, \Delta v_b\right)$

III. Check Eqn (1), adjust s and iterate until Eqn (1) is satisfied

- Solution by Analytical Approximation (0.5% error in inclination)

$$\begin{aligned}\Delta v^2 &= \Delta v_a^2 + \Delta v_b^2 + 2\Delta v_a v_b \\ &\approx \Delta v_a^2 + \Delta v_b^2 \\ &= v_{trans,a}^2 + v_{initial}^2 - 2v_{trans,a}v_{initial}\cos(s\Delta i) \\ &\quad + v_{trans,b}^2 + v_{final}^2 - 2v_{trans,b}v_{final}\cos((1-s)\Delta i)\end{aligned}$$

$$\frac{\partial(\Delta v^2)}{\partial s} \approx 2\Delta i v_{trans,a} v_{initial} \sin(s\Delta i) - 2\Delta i v_{trans,b} v_{final} \sin((1-s)\Delta i) = 0$$

$$\acute{Y} \quad \sin((1-s)\Delta i) = \sin(\Delta i - s\Delta i) = \sin\Delta i \cos(s\Delta i) - \cos\Delta i \sin(s\Delta i)$$

$$\Rightarrow v_{trans,a} v_{initial} \sin(s\Delta i) = v_{trans,b} v_{final} [\sin\Delta i \cos(s\Delta i) - \cos\Delta i \sin(s\Delta i)]$$

$$\Rightarrow v_{trans,a} v_{initial} \frac{\sin(s\Delta i)}{\cos(s\Delta i)} + v_{trans,b} v_{final} \cos\Delta i \frac{\sin(s\Delta i)}{\cos(s\Delta i)} = v_{trans,b} v_{final} \sin\Delta i$$

$$\Rightarrow \tan(s\Delta i)[v_{trans,a} v_{initial} + v_{trans,b} v_{final} \cos\Delta i] = v_{trans,b} v_{final} \sin\Delta i$$

Collecting terms and rearranging

$$\tan(s\Delta i) \approx \frac{\sin(\Delta i)}{\left[\frac{v_{initial} v_{trans,a}}{v_{final} v_{trans,b}} + \cos(\Delta i)\right]}, \qquad a_{transfer} = \frac{r_{initial} + r_{final}}{2}$$

$$\frac{v_{initial} v_{trans,a}}{v_{final} v_{trans,b}} = \frac{\sqrt{\frac{\mu}{r_{initial}}}\sqrt{\frac{2\mu}{r_{initial}} - \frac{\mu}{a_{trans}}}}{\sqrt{\frac{\mu}{r_{final}}}\sqrt{\frac{2\mu}{r_{final}} - \frac{\mu}{a_{trans}}}} = \frac{\sqrt{\frac{\mu}{r_{initial}}}\sqrt{2\mu}\sqrt{\frac{r_{final}}{r_{initial}(r_{initial} + r_{final})}}}{\sqrt{\frac{\mu}{r_{final}}}\sqrt{2\mu}\sqrt{\frac{r_{initial}}{r_{final}(r_{initial} + r_{final})}}} = \sqrt{\frac{r_{final}^3}{r_{initial}^3}} \equiv \sqrt{R^3}$$

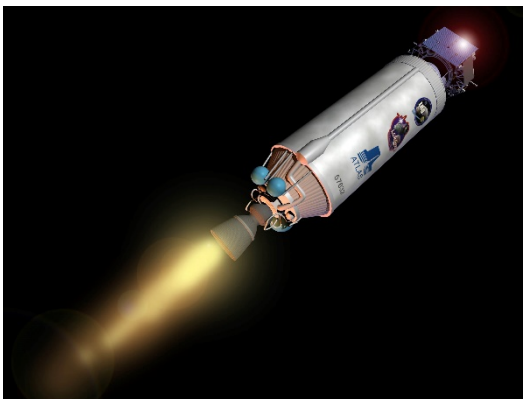
$$s \approx \frac{1}{\Delta i} \tan^{-1}\left\{\frac{\sin(\Delta i)}{R^{3/2} + \cos(\Delta i)}\right\}$$

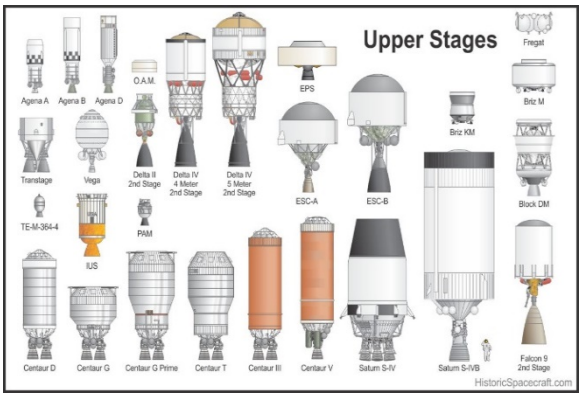
TABLE 6-3. Minimum Plane Changes and Δv for Common Orbital Transfers. The first line shows the nonoptimal change in velocity with the entire plane change on the second burn. Notice how each case places about 5% to 10% of the total inclination change into the first maneuver. To reverse the changes (going from higher altitude to lower), simply reverse all quantities. The optimal solutions are from the iteration method.

$r_{initial}$	r_{final}	$\Delta i°$	$\Delta i_a°$	$\Delta i_b°$	Δv_{total} m/s
6671.53 km	42,163.95 km	10.0	0.0	10.0	3945.42
(1.046 ER)	(6.6107 ER)	10.0	0.917	9.083	3940.90
6671.53 km	26,558.56 km	28.5	0.0	28.5	4121.20
(1.046 ER)	(4.164 ER)	28.5	3.305	25.195	4058.97
6671.53 km	42,163.95 km	45.0	0.0	45.0	4674.44
(1.046 ER)	(6.6107 ER)	45.0	2.751	42.249	4637.37

6.5.2 Fixed- Δv Maneuvering

- Motivation
  - For efficient orbit transfer, spacecraft usually uses upper stages (transfer booster)
  - Most of them have a fixed- Δv capacity, because they use solid propellant and cannot be turned on and off.



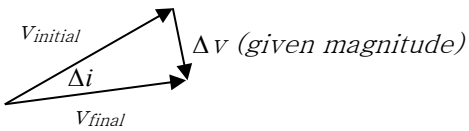


Centaur upper stage rocket firing with the Lunar CRater Observation and Sensing Satellite (LCROSS)

- Combined Maneuvers with two motors (stages)
  - The 1st burn completes a part of noncoplanar change in addition to changing the size and shape of the orbit
  - The 2nd burn at apogee completes the plane change and circularizes the final orbit

- Questions lies in how to compute:
  - The plane change Δi after Δv
  - The firing attitude (= payload angle = orientation of the burn= angle between $\vec{v}_{initial}$ and $\Delta\vec{v}$)

$$\gamma > 0 \ if \ \Delta i = i_{final} - i_{initial} > 0$$



$$\Delta v = \sqrt{v_{initial}^2 + v_{final}^2 - 2v_{initial}v_{final}\cos(\Delta i)}$$

$$\cos(\Delta i) = \frac{v_{initial}^2 + v_{final}^2 - (\Delta v)^2}{2v_{initial}v_{final}} \quad \text{(i)}$$

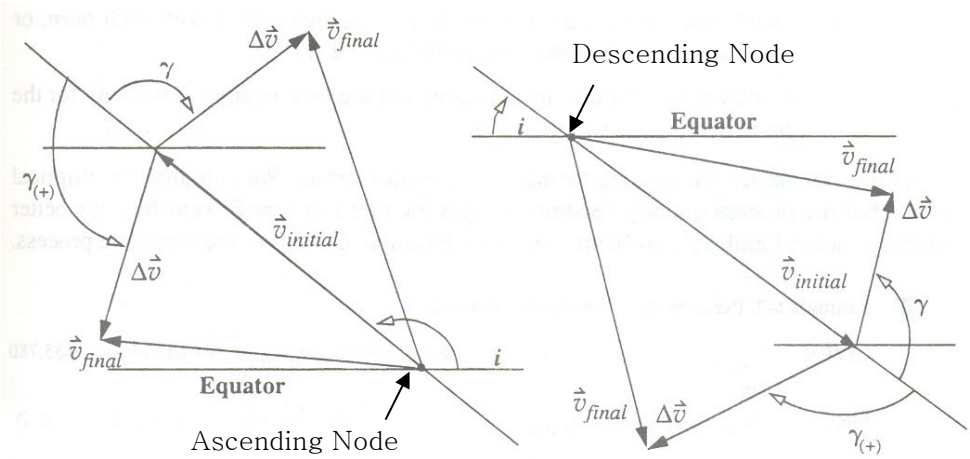


Figure 6-14. Geometry for Fixed-Δv Maneuvers. This figure includes changes to the retrograde ascending-node (left) and direct descending-node (right). I've shown *two* inclination changes (+ and –) for each case. Determine the proper sign for the payload angle, γ, based on the sign of the inclination change. [Positive γ values are shown with a (+).]

$$v_{final} = \sqrt{v_{initial}^2 + \Delta v^2 - 2v_{initial}\Delta v\, cos(180-\gamma)}$$

$$\Rightarrow cos(180-\gamma) = \frac{v_{initial}^2 + \Delta v^2 - v_{final}^2}{2v_{initial}\Delta v} \qquad \text{(ii)}$$

$$\Rightarrow cos(\gamma) = -\frac{v_{initial}^2 + \Delta v^2 - v_{final}^2}{2v_{initial}\Delta v}$$

- Assumption: Circular Orbits, Orbital Transfer at the Nodes
  Goal: Finding the firing attitude γ to achieve the required Δi
  - For the first burn, $v_{inital}= v_{inital}$, $v_{final}=v_{trans,a}$
  - For the second burn, $v_{inital}= v_{trans,b}$, $v_{final}= v_{final}$

- The first burn

  Given $v_{initial}$, $v_{trans,a}$, Δv_1

  Calculate γ_1, Δi_1
  Equation (1) and (2) are used to find γ_1 (from eq(ii)) and Δi_1 (from eq(i))

  $\Delta i<0, -180^o \le \gamma \le 0$

  $\Delta i>0, \quad 0 \le \gamma \le 180^o$

  The second burn can be analyzed in a similar way

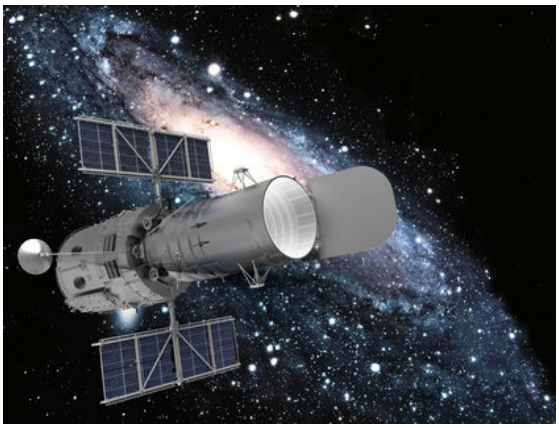
  Given $v_{trans,b}$, v_{final}, Δv_2,

  Calculate γ_2, Δi_2

6.6 Circular Rendezvous

- Circular Rendezvous as an application of the planar/non-planar orbital maneuvering
  - Need to Consider when the particular maneuvers occur
  - Only consider circular initial/final orbits with tangential burns for simplicity
    - More general approach will be studied in the frame of Lambert problem

  - Target refers to a reference point for times and angles; a desired endpoint for interceptor
  - Interceptor refers to a spacecraft to be placed at the desired endpoint (target)



Hubble Space Telescope mirror problem

6.6.1 Circular Coplanar Phasing

- Trailing Interceptor on the same Circular Orbit
  The interceptor trails the target by an angle θ (positive value in this note)

The mean motion n of the target

$$\omega_{tgt} = n = \sqrt{\frac{\mu}{a_{tgt}^3}}$$

$$\tau_{phase} = \Delta t = t_1 - t_0 = \frac{k_{tgt}(2\pi) - \theta}{\omega_{tgt}}, \quad k_{tgt} = 1, 2, \ldots$$

where $k_{tgt} = 1, 2, \ldots$ represents multiple revolutions of the target

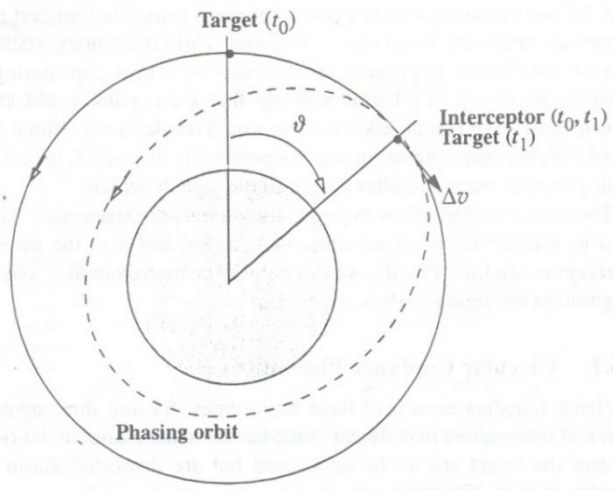


Figure 6-15. Circular Coplanar Rendezvous (Interceptor Trailing Target). Because the interceptor trails the target, the phase angle, ϑ, is negative. To rendezvous, the interceptor must move into a lower orbit with a shorter period to make up for the initial time displacement.

The phasing orbit consumes τ_{phase} as it accomplishes the rendezvous.

$$p = 2\pi\sqrt{\frac{a^3}{\mu}},\ p = \tau_{phase} \Rightarrow a_{phase} = \left[\mu\left(\frac{\tau_{phase}}{2\pi k_{int}}\right)^2\right]^{1/3},$$

$$\text{where } k_{int} = 1,2,\ldots \text{ multiple revolutions of the interceptor during } \tau_{phase}$$

$$r_p = 2a_{phase} - r_a \quad \text{where } r_a = r_{initial} = r_{target} \qquad \Leftarrow a = \frac{r_a + r_b}{2}$$

The first burn slows down the interceptor S/C to enter the phasing orbit.

$$\Delta v_1 = v_{int} - v_{phase} = \sqrt{\frac{\mu}{a_{tgt}}} - \sqrt{\frac{2\mu}{a_{tgt}} - \frac{\mu}{a_{phase}}}$$

The second burn speeds up the interceptor S/C to reenter the target orbit.

$$\Delta v_2 = v_{int} - v_{phase} = \sqrt{\frac{\mu}{a_{tgt}}} - \sqrt{\frac{2\mu}{a_{tgt}} - \frac{\mu}{a_{phase}}}$$

The initial and final burns have the same magnitude but opposite directions.

$$\text{Total } \Delta v = \Delta v_1 + \Delta v_2 = 2\left|v_{int} - v_{phase}\right| = 2\left|\sqrt{\frac{\mu}{a_{tgt}}} - \sqrt{\frac{2\mu}{a_{tgt}} - \frac{\mu}{a_{phase}}}\right|$$

$$\text{Increase } \tau_{phase} \Rightarrow \text{Increase } a_{phase} \Rightarrow \text{Decrease } \Delta v$$

- Leading Interceptor on the same Circular Orbit

  The interceptor leads the target by an angle θ (positive value in this note)

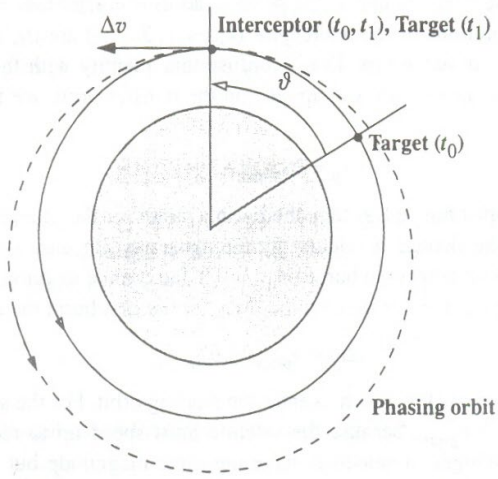


Figure 6-16. Circular Coplanar Rendezvous (Interceptor Leading Target). When the interceptor leads the target, the phase angle is positive, and the phasing orbit must be longer than the target orbit to permit the target satellite to *catch up*.

To rendezvous, the target needs to travel one complete orbit plus the phase angle θ, whle the interceptor completes one revolution on the phasing orbit.

$$\omega_{tgt} = n = \sqrt{\frac{\mu}{a_{tgt}^3}}$$

$$\tau_{phase} = \Delta t = t_1 - t_0 = \frac{k_{tgt} 2\pi + \theta}{\omega_{tgt}}, \qquad k_{tgt} = 1,2,...$$

$$a_{phase} = \left[\mu \left(\frac{\tau_{phase}}{k_{int}(2\pi)} \right)^2 \right]^{1/3}, \quad k_{int} = 1,2,... \quad \leftarrow \quad P = 2\pi\sqrt{\frac{a^3}{\mu}}$$

The initial and final burns have the same magnitude but opposite directions.

$$\text{Total} \quad \Delta v = \Delta v_1 + \Delta v_2 = 2\left| v_{phase} - v_{int} \right| = 2\left| \sqrt{\frac{2\mu}{a_{tgt}} - \frac{\mu}{a_{phase}}} - \sqrt{\frac{\mu}{a_{tgt}}} \right|$$

$$Increase \;\; \tau_{phase} \Rightarrow increase \;\; a_{phase} \Rightarrow \;\; increase \;\; \Delta v$$

- Target and Interceptor on Separate, Coplanar Circular Orbits

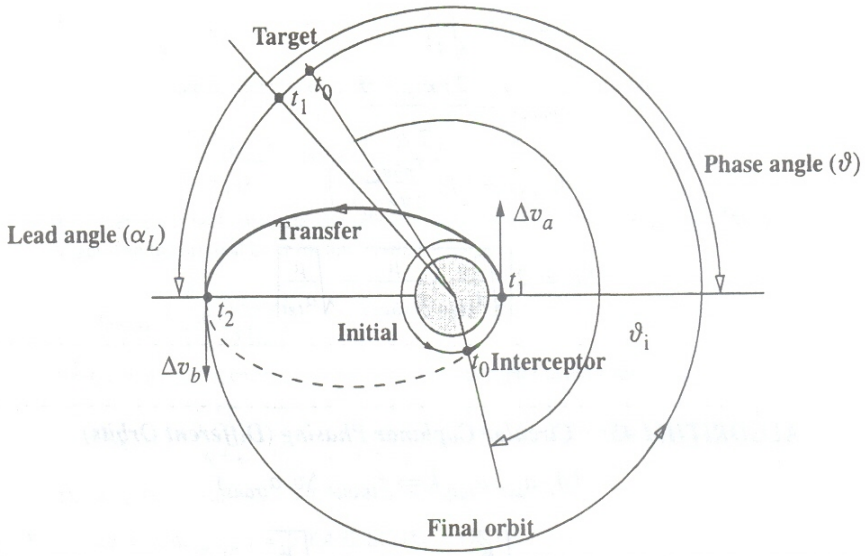


Figure 6-17. Circular Coplanar Rendezvous (Different Orbits). Whenever the satellites are in different orbits, we must calculate a lead angle (and therefore a phase angle). Also, the initial separation of the satellites (at t_0) usually differs from the required phasing for transfer, thus the symbol for the initial phase, ϑ_i. The final intercept occurs at t_2.

The interceptor minimizes ΔV by using a Hohmann transfer to rendezvous with the target. The time for the transfer orbit:

$$\tau_{trans} = \pi\sqrt{\frac{a_{trans}^3}{\mu}}, \qquad a_{trans} = \frac{r_{initial} + r_{target}}{2}$$

Find the lead angle:

$$\alpha_L = \omega_{tgt}\tau_{trans} \quad \text{where } \omega_{tgt} = n = \sqrt{\frac{\mu}{a_{tgt}^3}}$$

Find the required phase angle:

$$\theta = \pi - \alpha_L$$

The waiting time, τ_{wait} is for the interceptor and target to be in the correct positions, which depends on the phase angle separations and the relative velocities of the two spacecraft

$$\tau_{wait} = \frac{(\theta_i - \theta) + 2\pi k}{(\omega_{int} - \omega_{tgt})}$$

where θ_i *is the initial phase angle*

ω_{int} *is the interceptor angular velocity,* $\omega_{int} = \sqrt{\frac{\mu}{a_{int}^3}}$

$k = 0, 1, 2, \ldots$

If the two spacecraft are on the orbits of similar size, τ_{wait} is large.

Total $\Delta v = |\Delta v_a| + |\Delta v_b|$ by Hohamann transfer

$$\Delta v_a = \left| \sqrt{\frac{2\mu}{a_{int}} - \frac{\mu}{a_{trans}}} - \sqrt{\frac{\mu}{a_{int}}} \right|, \qquad \Delta v_b = \left| \sqrt{\frac{\mu}{a_{final}}} - \sqrt{\frac{2\mu}{a_{final}} - \frac{\mu}{a_{trans}}} \right|$$

6.6.2 Circular Non-coplanar Phasing

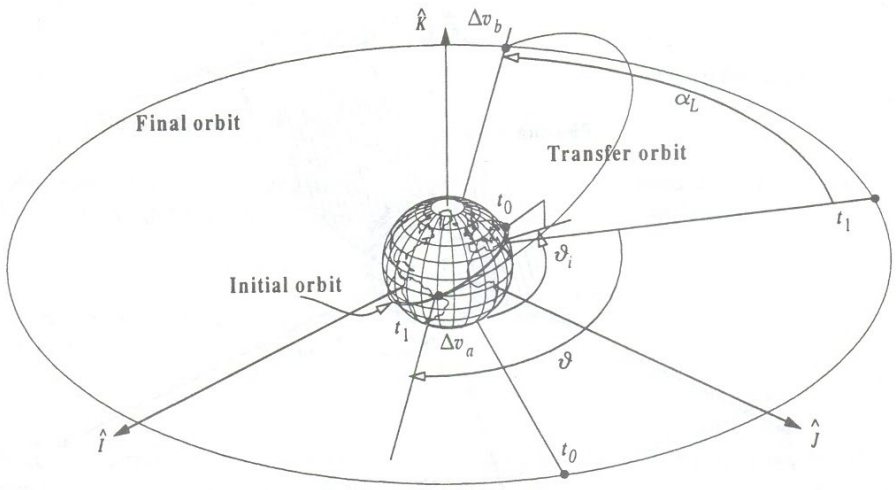


Figure 6-18. Geometry for a Circular, Noncoplanar, Orbit Transfer. This transfer combines the Hohmann transfer from LEO to GEO with the change in inclination plane. The maneuver can't start until the interceptor is at a node, t_1, and the target, t_1, is at the required lead angle. Because both satellites begin at t_0, this event may require an extremely long wait time. Notice the convention used in measuring angles *from* the target satellite. When drawn to scale, the LEO orbit appears almost on the Earth.

- Not always timely to wait for the natural motion of satellites to place the interceptor at **<u>the node</u>** and the target at the correct phase angle necessary for the non-coplanar intercept
  - Can use phasing orbits to help place the interceptor at a node at the proper time
  - Can change the velocity to enter the phasing orbit so it becomes a part of total Δv $\Rightarrow$ No waste in velocity
  - Assume circular orbits, Hohmann transfer

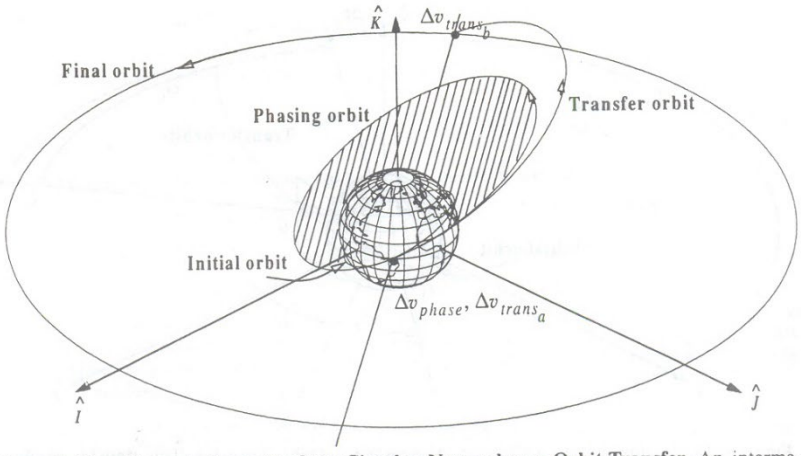


Figure 6-19. Alternate Geometry for a Circular, Noncoplanar, Orbit Transfer. An intermediate phasing orbit is useful to permit a quicker time to rendezvous. The size of the phasing orbit is calculated so the satellite returns to the node in time for the transfer into the final orbit. Both burns (Δv_{phase} and Δv_{trans_a}) occur at the same point.

- The period of the phasing orbit is precisely designed so that the interceptor travels from time t_p and arrives back at the node at the same time the phase angle is correct to start the Hohmann transfer

Semimajor Axis of the transfer orbit

$$a_{\text{trans}} = \frac{a_{\text{initial}} + a_{\text{final}}}{2}$$

Time of flight for the transfer

$$\tau_{\text{trans}} = \pi\sqrt{\frac{a_{\text{trans}}^3}{\mu}}$$

Lead angle

$$\alpha_L = \omega_{\text{tgt}}\tau_{\text{trans}} \quad \text{where} \quad \omega_{\text{tgt}} = n = \sqrt{\frac{\mu}{a_{tgt}^3}}$$

$\Delta\theta_{\text{int}}$: Angle the interceptor must travel until reaching perigee of the transfer orbit

Δt_{node} : Amount of time required for $\Delta\theta_{\text{int}}$

$$\Delta t_{\text{node}} = \Delta\theta_{\text{int}} / \omega_{\text{int}}$$

The angular distance (longitude) the target has traveled from its initial location

$$\lambda_{\text{true,tgt1}} = \lambda_{\text{true,tgt0}} + \omega_{\text{tgt}}\Delta t_{\text{node}}$$

True longitude for interceptor at t_1

$\lambda_{true,int1}=\Omega+u_{int}$, *where* Ω*: longitude of ascending node,*

$u_{int}=w+v=0$ *at the ascending node*

New phase angle at t_1

$$\theta_{new} = \lambda_{true,\text{int}1} - \lambda_{true,tgt1} \qquad \text{(negative number for this case)}$$

New lead angle

$$\alpha_{new} = \pi + \theta_{new}$$

The phasing orbit must account for the time between the lead angles

$$p_{phase}=\frac{(\alpha_{new}-\alpha_L)+2\pi k_{tgt}}{\omega_{tgt}}, \qquad k_{tgt}=0,1,2,...$$

Knowing the period of phasing orbit, calculate the semimajor axis of the phasing orbit.

$$a_{phase}=\left[\mu\left(\frac{p_{phase}}{k_{int}(2\pi)}\right)^2\right]^{1/3}, \qquad k_{int}=1,2,... \quad \Leftarrow P=2\pi\sqrt{\frac{a^3}{\mu}}$$

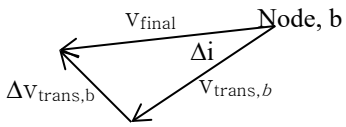
The velocity change to enter the phasing orbit

$$\Delta v_{phase} = \left|\sqrt{\frac{2\mu}{a_{int}}-\frac{\mu}{a_{phase}}}-\sqrt{\frac{\mu}{a_{int}}}\right|$$

The velocity change to enter the transfer orbit

$$\Delta v_{trans,a} = \left|\sqrt{\frac{2\mu}{a_{int}}-\frac{\mu}{a_{trans}}}-\sqrt{\frac{2\mu}{a_{int}}-\frac{\mu}{a_{phase}}}\right|$$

The velocity change to complete the transfer to target orbit.



$$v_{trans,b}=\sqrt{\frac{2\mu}{a_{tgt}}-\frac{\mu}{a_{trans}}}, \quad v_{final}=\sqrt{\frac{\mu}{a_{tgt}}}$$

$$\Delta v_{trans,b}=\sqrt{v_{trans,b}^2+v_{final}^2-2v_{trans,b}v_{final}\cos(\Delta i)}$$

Total $\Delta v=\Delta v_{phase}+\Delta v_{trans,a}+\Delta v_{trans,b}$

$$\tau_{total}=\Delta t_{node}+2\pi\sqrt{\frac{a_{phase}^3}{\mu}}+\tau_{trans}, \qquad \tau_{trans} = \pi\sqrt{\frac{a_{trans}^3}{\mu}}$$